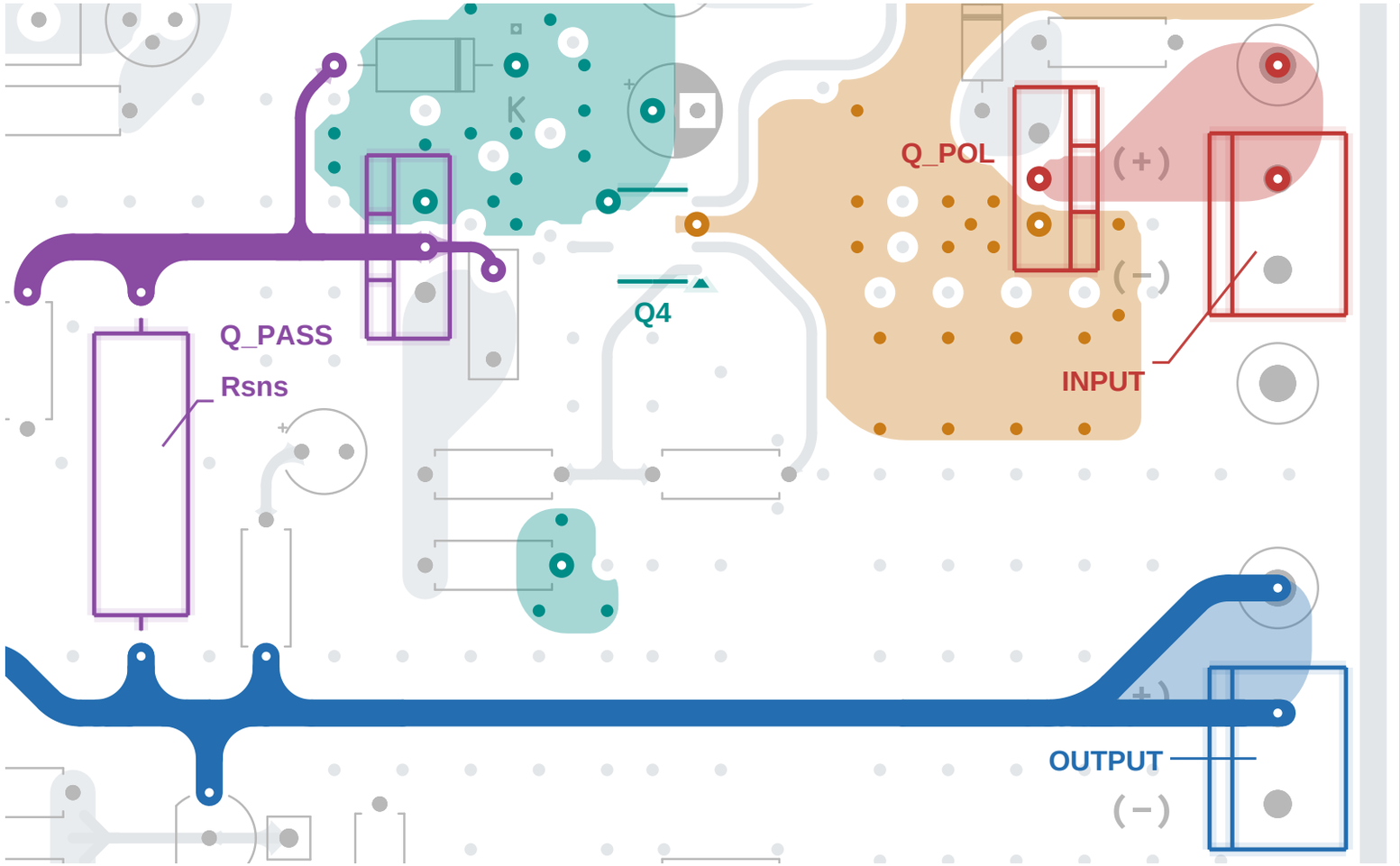
Input-to-output power path

Actual KiCad copper + component highlights | Top copper and front silkscreen



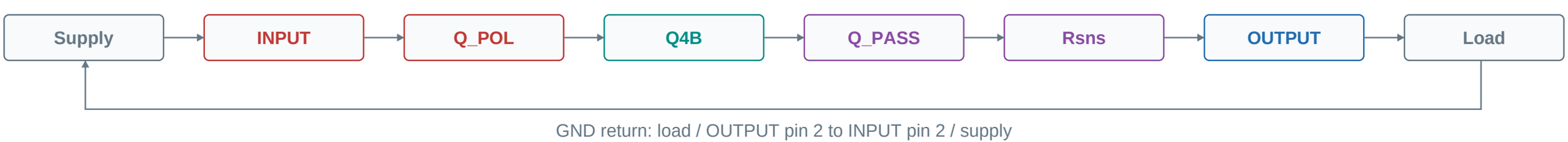
Input and polarity protection
VCC enters Q_POL at pin 2.

Protected supply
VEE joins Q_POL pin 3 to Q4B pin 3.

Startup output
Q4B pins 3-5 feed the pass emitter. Q4 is a dual-device package.

Pass collector
Q_PASS pin 2 reaches Rsns pin 1.

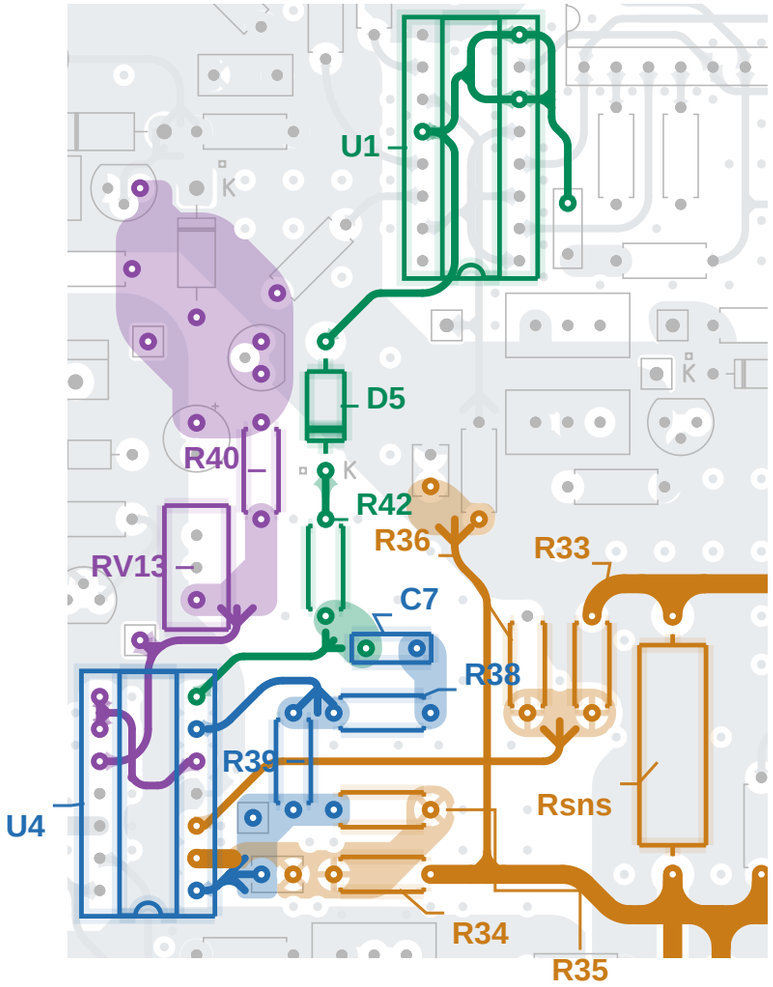
Load output
Rsns pin 2 reaches OUTPUT pin 1. The load is external.



Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.

Current sensing and limiting

Actual KiCad copper + component highlights | Top-side projection of both outer layers



A shared color may span series components. Their pads and copper breaks remain distinct.

Differential sensing

R33 / R34 pick up opposite sides of Rsns. R35 / R36 complete the amplifier network.

Amplified signal

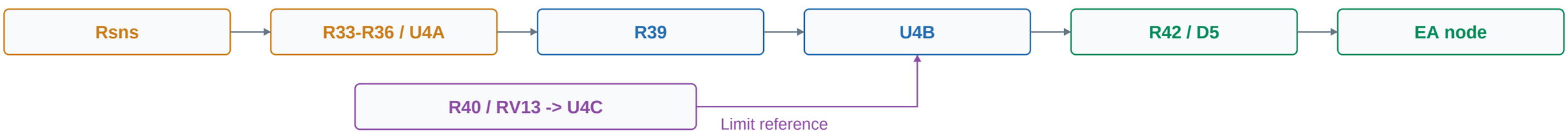
U4A output at pin 1 reaches U4B through R39. U4 contains several amplifier sections.

Adjustable limit

R40 / RV13 set the threshold; U4C buffers it for U4B.

Limiting action

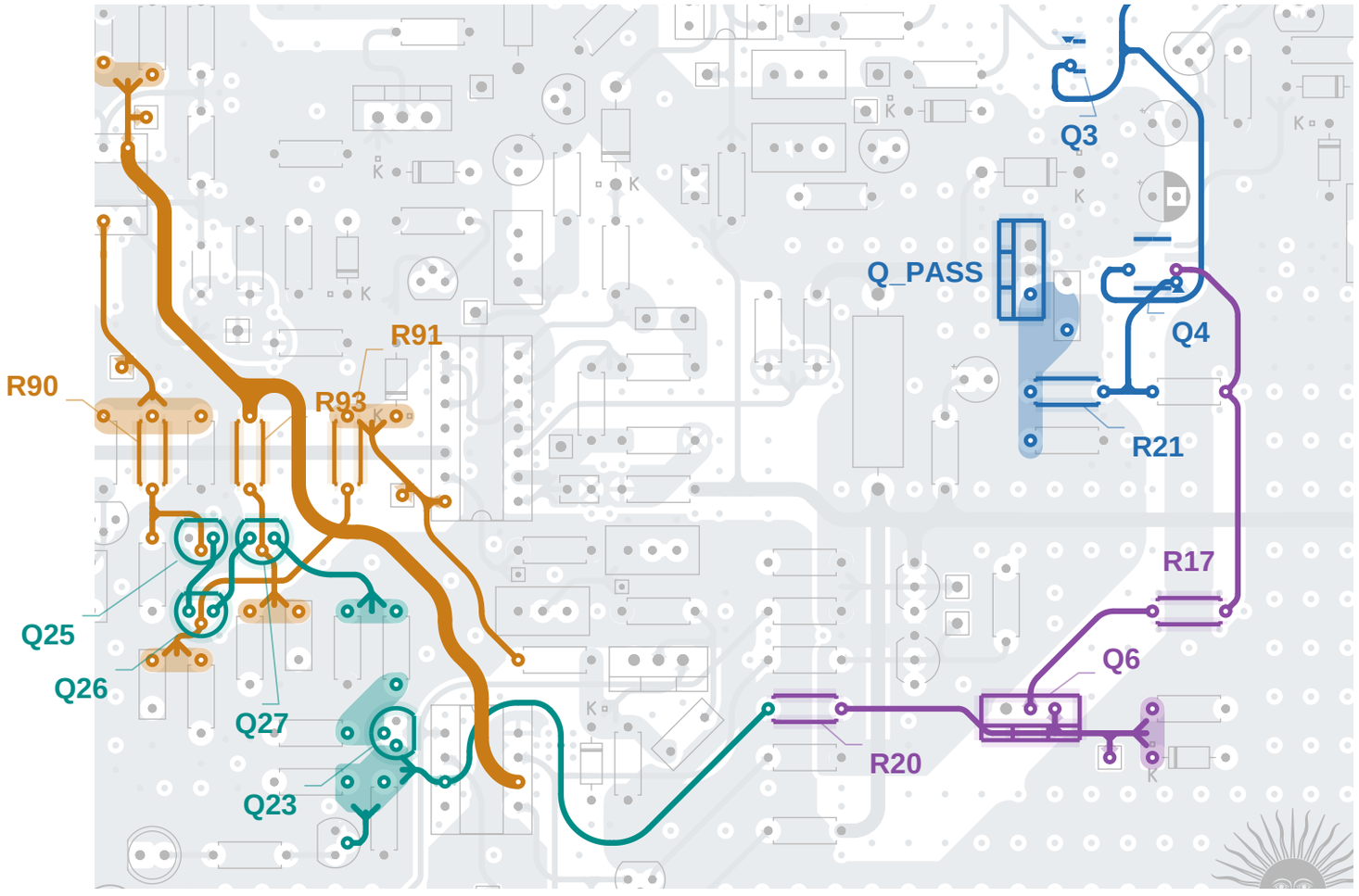
U4B output reaches the error-amplifier node through R42 and D5.



Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.

Enable logic and pass-drive switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers



Q4A and Q4B share the Q4 footprint. This view highlights the enable section Q4A.

Permission inputs

PCB nets 1 (UVLO), 2 (OVLO), and LDO2 (delay) drive the series NPN chain.

Discrete AND function

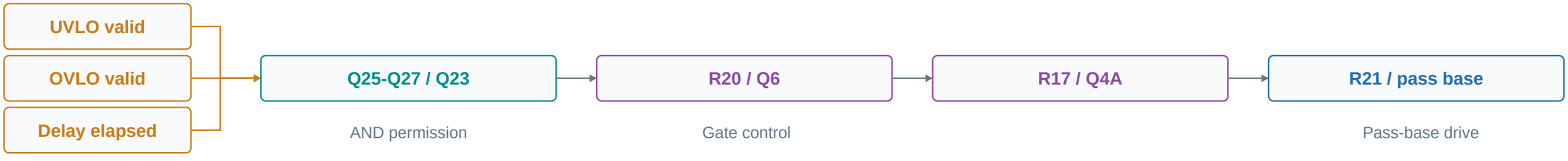
Q25-Q27 and the inverter Q23 generate the TTL enable node.

Gate control

TTL drives Q6 through R20. Q6 controls Q4A through R17.

Pass-base drive

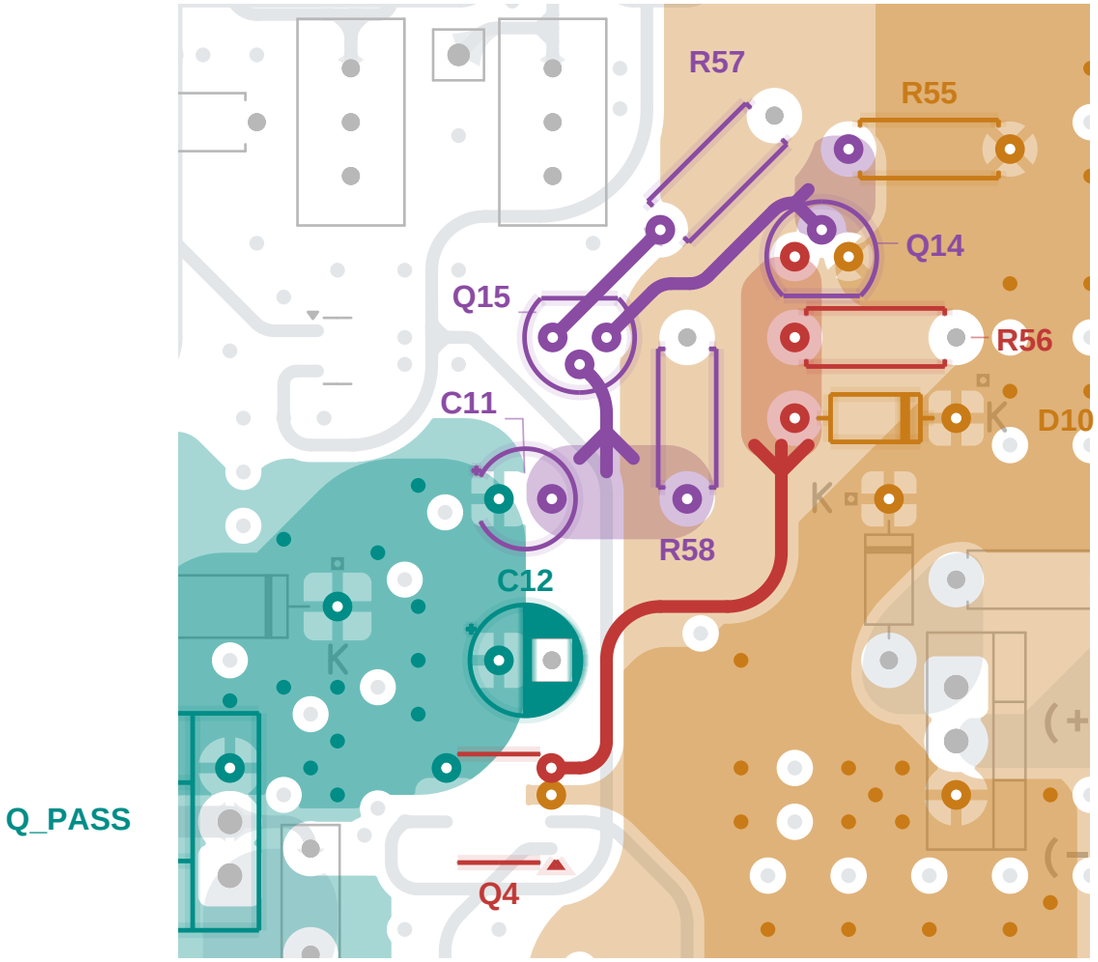
Q4A connects the Q3 output-stage path through R21 to the pass base.



Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.

Startup ramp and series switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers



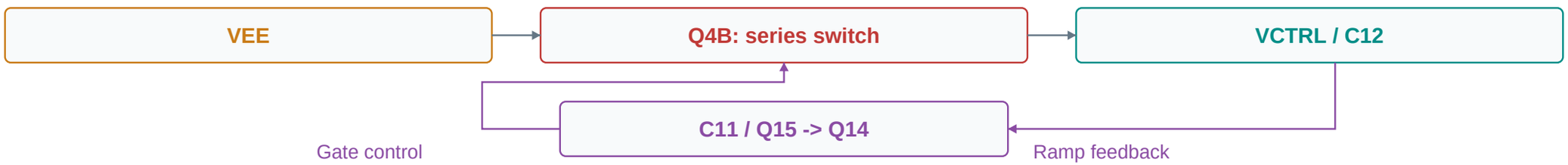
Protected supply
VEE supplies the startup stage and Q4B pin 3.

Ramp feedback
C11 couples changes of VCTRL into the Q15 / Q14 control network.

Switch gate
The Q14-controlled node drives Q4B gate pin 4; D10 connects it to VEE.

Ramped supply
Q4B pin 5 feeds VCTRL, C12 and the pass emitter. C12 is 10 uF in the PCB.

Q4B is the startup section of the shared Q4 package. The enable section Q4A is a separate path.



Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.